

Problem Faced

High risk from
undetected fatigue, toxic
gas, and vital sign
detection

01

No real-time integrated
monitoring of health,
fatigue, and gas
exposure

02

Poor communication
with control
center/medical team in
remote areas.

03

Difficulty monitoring
and protecting rescuers
in hazardous,
inaccessible zones

04

Limited situational
awareness and
coordination in multi-
risk environments.

05

Solution

Continuous monitoring of vital
and environmental conditions

Integration with command center
dashboards

Seamless connectivity for
remote health supervision

Waterproof and shock-resistant
design for field safety

Proactive alerts to prevent
responder burnout or collapse

Supports team-level tracking to
coordinate rescues better

Compatible with harsh or
chemical environments

Ensures timely intervention and
responder safety

Need of our device



Industrial and
disaster risks
demand
continuous, on-
body safety
monitoring in real
time.

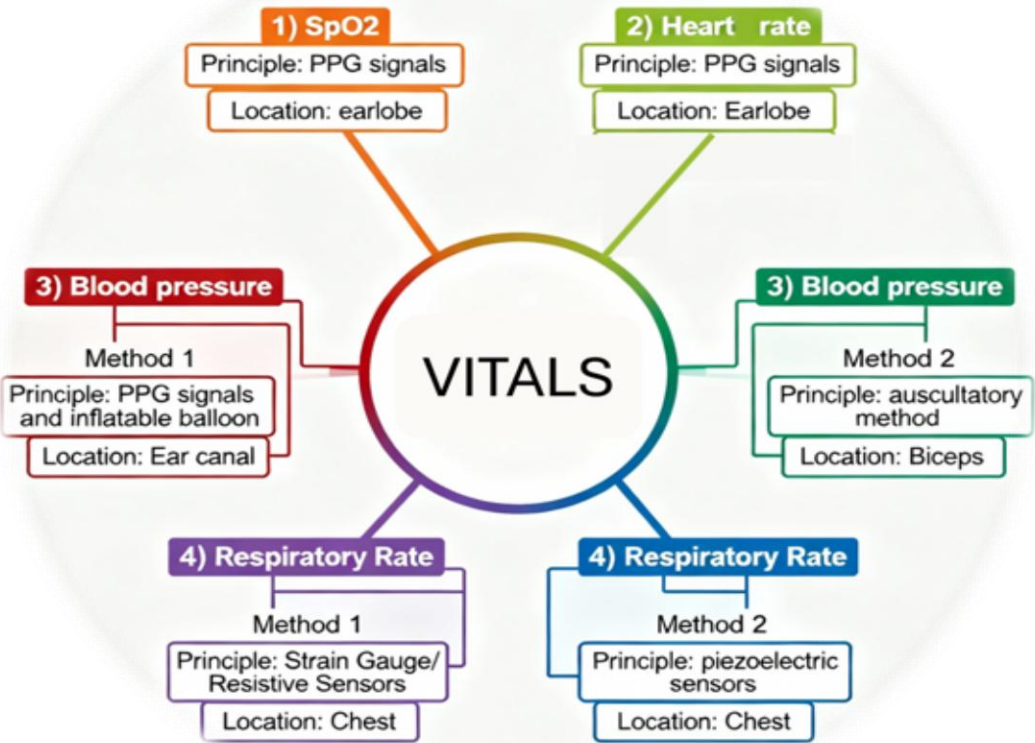
Require reliable
monitoring that
works in harsh
conditions.



Seamless, live
data sharing
enables rapid
response and
oversight.

Delivers
peace of mind
for teams,
supervisors.



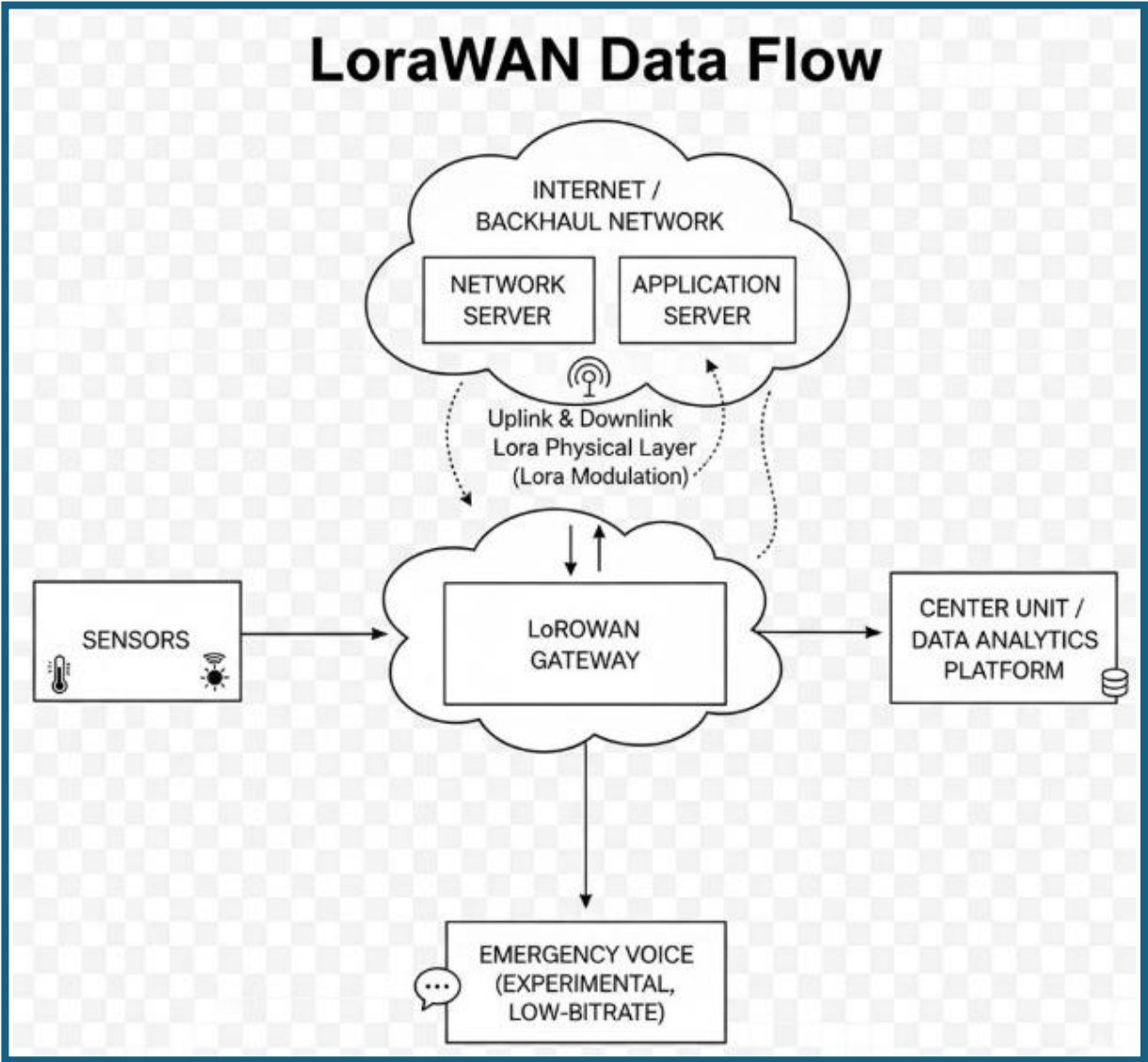


NOTE:

The location of the sensors and the methods for measuring the vitals can vary based on the rescue environment. Thus our device is **adaptable** to the rescue environments and conditions.

FATIGUE:

Using these vital parameters fatigue monitoring can be modelled.



TECHNICAL APPROACH

O_2 (Mandatory)

- Mechanism: Electrochemical sensor
- Location: Near breathing area of rescuer and on shoulder of them

CH_3

- Mechanism: Infrared sensor(accurate) or NDIR sensor technology, semiconducting metal oxide
- Location: Helmet Chin/Lower Face Area, Collar, near Oxygen tank

$C_3 H_8 / C_4 H_{10}$

- Mechanism: Infrared sensor(accurate) or NDIR sensor technology, semiconducting metal oxide-
- Location: Helmet Chin/Lower Face Area, Collar, near Oxygen tank

H_2S

- Mechanism: Electrochemical and solid-state sensors not Waterproof
- Location: Low as possible at waist, near Oxygen tank

CO

- Mechanism: Electrochemical sensor
- Location: -> Helmet Chin/Lower Face Area, Collar, near Oxygen tank

NH_3

- Mechanism: Electrochemical sensor
- Location: Helmet Chin/Lower Face Area, Collar, near Oxygen tank

Cl_2

- Mechanism: Electrochemical sensor
- Location: Helmet chin, Shoulder/collar

CO_2

- Mechanism: Electrochemical sensor
- Location: Helmet Chin/Lower Face Area, Collar, near Oxygen tank

SO_2

- Mechanism: Electrochemical sensor
- Location: Helmet Chin/Lower Face Area, Collar, near Oxygen tank

HCN

- Mechanism: Electrochemical sensor
- Location: Helmet Chin/Lower Face Area, Collar, near Oxygen tank

NO_2

- Mechanism: Electrochemical sensor
- Location: Helmet Chin/Lower Face Area, Collar, near Oxygen tank

Gas Detection & mechanism

These sensors will be used as per the Rescue environment and hazardous conditions

Challenges and Risks

Accuracy and Reliability:

Sensor readings may be affected by noise, temperature, or user motion, causing false or missed alerts.

Battery Life: Continuous monitoring and data transmission drain power quickly.

Connectivity issues:

Weak or unstable signals can interrupt monitoring and delays alerts.

Data Privacy: Health data must be securely transmitted and stored.

Strategies for Overcoming Challenges

01

Use AES encryption for secure data and communication.

02

Reduce power use with efficient code and low-power modes.

03

Validate system accuracy with healthcare experts.

04

Test sensors in diverse real-world environmental conditions or diverse hazardous situation.

05

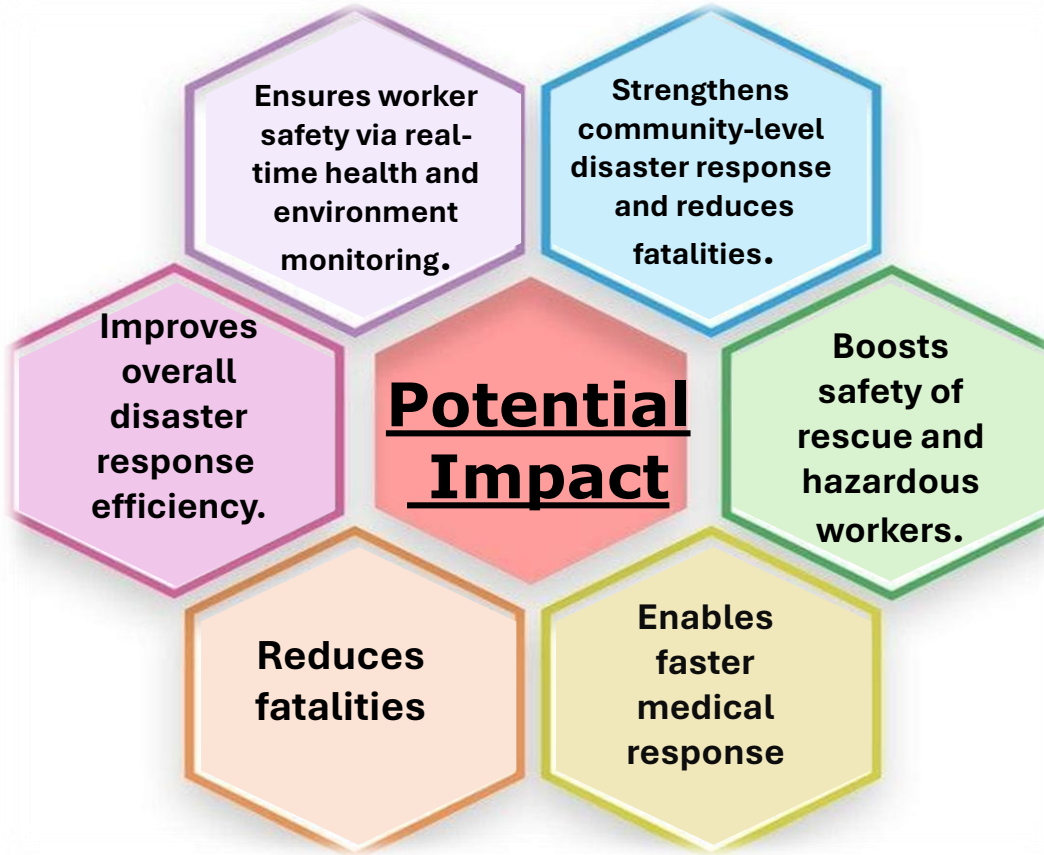
Implement multi-layer emergency verification before final activation.

06

Ensure reliable connectivity/communication with redundant methods



IMPACT AND BENEFITS



Benefits:



Promotes inclusive healthcare for rescuers.



Enhances safety through real-time health monitoring.



Strengthens communication with control center/medical teams.



Builds trust in tech-based healthcare.

RESEARCH AND REFERENCES

- <https://www.tandfonline.com/doi/full/10.2147/MDER.S47319>
- <https://www.mdpi.com/1424-8220/18/3/844>
- <https://dl.acm.org/doi/abs/10.1145/3300061.3345454>
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